Smart Puck

Assembly guide · cordless build · November 2026



This guide takes you from printed parts to a working, battery-powered Smart Puck: a 1.28" ESP32-S3 round knob display sitting on a printed tub and lid, with a LiPo battery, a USB-C charge board and a rocker switch inside. Plan on about an hour, most of it soldering.

1. Parts and tools

#	Part	Qty	Notes
A	1.28" ESP32-S3 round knob touch display	1	ESP32-S3, 240×240 round touchscreen, dial and press built in. Comes with its 4-pin-to-USB cable; keep it for reflashing.
B	Printed tub (puck_tub.stl)	1	Bottom half: battery strips, charge-board rails, rocker notch, screw counterbores.
C	Printed lid (puck_lid.stl)	1	Top half: 46.2 mm knob pocket, 18 mm deep, tilted 7.5° toward you.
D	103450 LiPo, 3.7 V 2000 mAh, protected	1	About 50 × 34 × 10 mm.
E	USB-C charge + 5 V boost board	1	26.1 mm long. Charges the battery and outputs 5 V. USB-C is power only.
F	Black 2-pin snap-in mini rocker switch	1	Lip 20.76 × 12.6 mm, about 12 mm deep. Rated 1 A or more.
G	MX1.25 4-pin plug wires	1	Plugs into the knob's "USB IN (5V)" socket.
H	M3 × 16 socket-head screws	4	Join the tub to the lid.
I	Soldering iron with heat-set insert tool	1	Optional. For M3 heat-set inserts in the lid instead of threading the screws into plastic.
J	Black PLA+ and grey PLA+	1 + 1	Black for the lid, grey for the tub.
K	10 mm round rubber feet	4	Optional. Fit the recesses under the tub.

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Tools and supplies: soldering iron and rosin-core solder, heat-shrink or electrical tape, flush cutters and wire strippers, 2.5 mm hex key, double-sided foam tape, and a multimeter (strongly recommended).

> **Important.** LiPo cells can catch fire if shorted. Cut and strip one battery wire at a time, never let the bare ends touch, and keep tape over any exposed lead until it is soldered.

2. Print the tub and lid

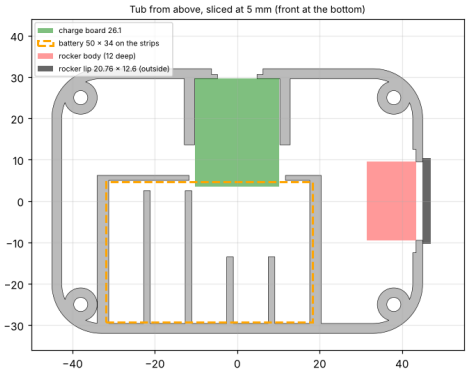
	Tub	Lid
File	hardware/stl/puck_tub.stl	hardware/stl/puck_lid.stl
Orientation	Floor down, as exported	Flat split face down, as exported
Supports	None	None (solid under the knob pocket)
Material	Grey PLA+	Black PLA+
Settings	0.2 mm layers, 3 walls, 15–20% infill	0.2 mm layers, 3 walls, 15–20% infill

Finished case: 90 × 64 mm, 38.5 mm tall at the centre. The tub and lid split 13 mm up from the floor. The reference build was printed on a Bambu Lab A1 mini.

> **Tip.** The 46.2 mm knob pocket was tuned on an A1 mini for a snug press fit. If your printer prints holes tight or loose, print hardware/stl/fit_test_knob_hole_46.0-46.6.stl first and pick the hole that grips (1 to 4 dots = 46.0 to 46.6 mm).

3. Know the layout

Looking down into the tub with the front edge toward you:

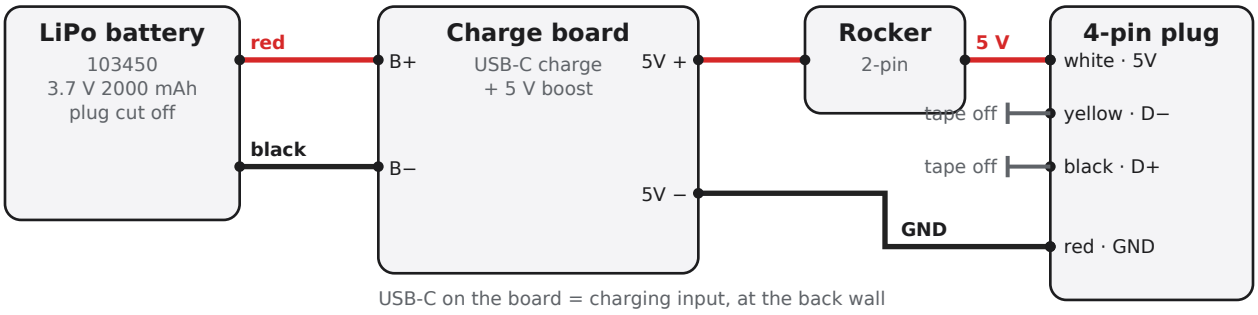


Tub from above. Green: charge board in its rails at the back wall, USB-C facing out. Orange dashed: battery on its strips, front-left. Red and dark grey: the rocker in the notch in the right end wall.

Feature	Where
Charge-board rails	Back wall, centred. The board slides down into two grooved rails and rides 1.2 mm above the floor so its underside solder clears.
Battery strips	Front-left. 5.7 mm tall so the battery sits just above the charge board and overhangs its front edge.
Rocker notch	Top of the right end wall, 19.0 × 10.8 mm. The lid closes the top of the notch.
Cord opening	D-shaped hole in the back of the lid's knob pocket. The 4-pin pigtail comes up through it.
Screws	Four M3 × 16, from underneath the tub up into the lid's 2.6 mm pilot holes (or optional heat-set inserts).

4. Wire it up

Everything is on the 5 V side: the battery feeds the charge board, the board's 5 V output goes through the rocker to the knob. The data pins are not used.



Important. Match the 4-pin plug by pin position, not by wire colour. On the reference pigtail, as plugged into the knob, **white is 5 V and red is GND**, the opposite of what the colours suggest. Getting this backwards reverses the polarity. Before soldering, plug the pigtail into the knob and trace which wire lands on the pin labelled 5V.

Battery to charge board

1. Cut the plug off the battery. Strip and tin the red wire only, then solder it to the board's **B+** pad. Tape it.
2. Now strip and tin the black wire and solder it to **B−**.
3. Plug a USB-C charger into the board. Its charge LEDs should light. Unplug it again.

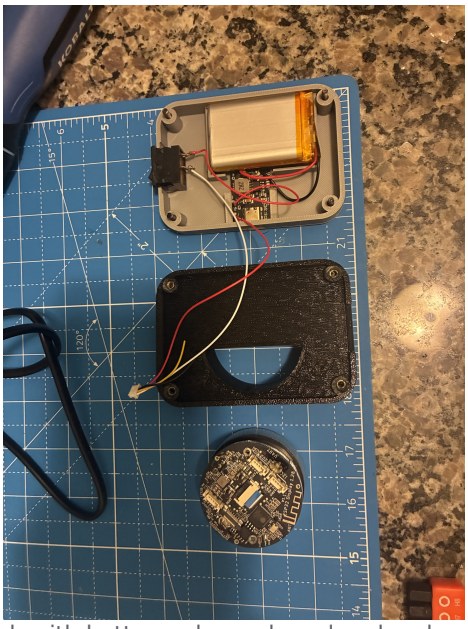
Board to rocker to knob

4. Solder a short red wire from the board's **5V +** output pad to one rocker terminal.
5. Solder the pigtail's **5 V wire** (white on the reference build) to the other rocker terminal.
6. Solder the pigtail's **GND wire** (red on the reference build) to the board's **5V −** pad.
7. Insulate the two data wires (yellow and black on the reference build) separately with tape or heat-shrink. They stay unconnected.
8. Cover every joint with heat-shrink or tape. Leave enough pigtail (about 10 cm) to reach up through the lid to the knob.

Test before connecting the knob

9. Flip the rocker on. With a multimeter on DC volts, put the red probe on the pigtail's 5 V pin and the black probe on its GND pin. You should read about **+5 V**.
10. If you read **−5 V**, the pigtail wires are swapped. Fix them now.
11. Flip the rocker off and confirm the reading drops to 0.

5. Put it together



Reference build before closing: tub with battery, charge board and rocker wired in (top), lid with the pigtail fed through the cord opening (middle), and the knob display face down (bottom).

1. **Heat-set inserts (optional).** The screws thread straight into the lid's 2.6 mm pilot holes, so inserts aren't required. If you want them, open the four holes to the size your inserts call for (usually about 4 mm), then press an M3 insert straight down into each with the soldering iron's insert tip until it sits flush. Let them cool before going on.
2. **Charge board.** Lower the board into the rails at the back wall, components up and USB-C facing out, and slide it down until the connector lines up with the slot.
3. **Rocker.** Drop the rocker into the notch in the right end wall from above, lip on the outside. Its tabs barely flex, so it goes in from the top rather than pushing through. It stands about 2 mm above the rim; the lid has a notch for that.
4. **Battery.** Lay the battery flat on the strips at the front-left, with its back edge over the charge board. Hold it with a strip of foam tape. Tuck the wires so none sit on top of the screw columns.
5. **Pigtail.** Feed the 4-pin plug up from inside the lid, through the D-shaped cord opening in the back of the knob pocket.
6. **Knob.** Plug the pigtail into the knob's "USB IN (5V)" socket. Turn the knob so the plug sits over the cord opening, then press the knob straight down into the pocket until it seats on the floor. It is a press fit.
7. **Close.** Set the lid on the tub, rocker end to rocker end, checking no wire is pinched at the seam. Drive the four M3 × 16 screws up from underneath into the lid. Snug, not tight: without inserts they cut their own thread in plastic.
8. **Feet.** Stick the rubber feet into the four recesses under the tub.

6. Power on and charge



Rocker switch on the right end of the finished puck.

1. Flip the rocker on. The screen should light and run the startup animation into the clock face.
2. To charge, plug any 5 V USB-C charger into the port at the back. Charging works with the rocker on or off.
3. The charge board turns its output off when nothing draws power, so the puck has no deep-sleep mode: off is off.

Updating firmware

The charge board passes power only, so the USB-C port can't flash the puck. Update over Wi-Fi (OTA) once the puck is on your network, or lift the knob out of the lid, unplug the pigtail and connect the knob's original cable to your computer.

Troubleshooting

Symptom	Check
Nothing at all	Is the battery charged? Is the rocker wired in the 5 V + line? Measure 5 V at the pigtail with the rocker on.
Board LEDs light, knob stays dark	Pigtail polarity. Re-check 5 V and GND by pin position (section 4).
Charge board won't slide in	Solder blobs underneath. Trim them flush; the rails leave 1.2 mm of clearance.
Knob spins in the pocket	Your printer runs holes large. Reprint the lid with a smaller pocket (see the fit test in section 2).
Lid won't close	A wire is on a screw column or the rocker isn't fully down in its notch.